Statistical Evaluation of Global Average Temperature and Sea Surface Temperature

A hypothesis testing activity submitted by

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1. About the Dataset

The dataset used in this analysis contains climate-related measurements collected from global environmental observations. In particular, the variables analyzed in this study include global average temperature and sea surface temperature both measured in Celsius (°C). These variables are commonly used in climate science to analyze trends in global warming and ocean temperature behavior.

source: https://www.kaggle.com/datasets/algozee/climate-cahnge

2. Statistical Question

Is the global average temperature significantly different from the sea surface temperature based on the collected observations?

3. Hypotheses

$H_o: \mu_{global\ temp} = \mu_{sea\ temp}$

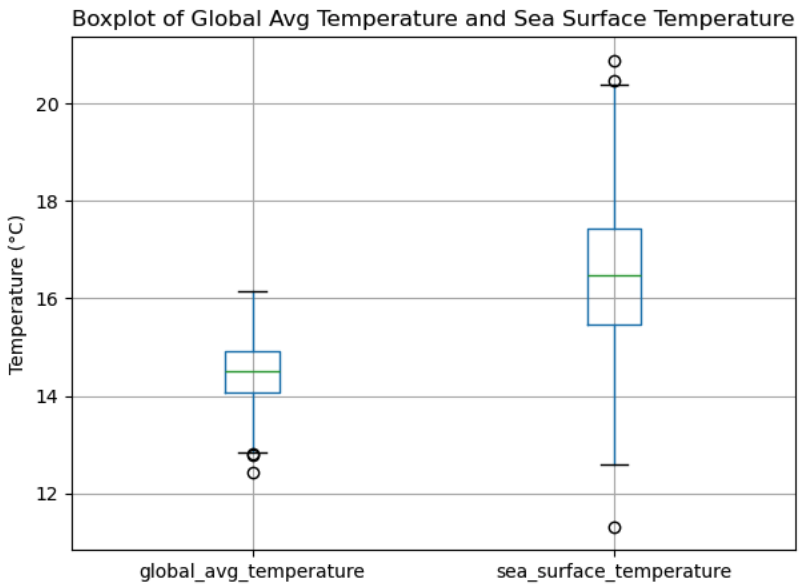
$H_a: \mu_{global\ temp} > \mu_{sea\ temp}$

4. Test Statistic

4.1 Summary of statistics

···	# global_avg_temperature	# sea_surface_temperature
count	1200.0	1200.0
mean	14.492849999999999	16.472383333333333
std	0.5945880843000982	1.4704983137691412
min	12.42	11.32
25%	14.08	15.4575
50%	14.51	16.46
75%	14.91	17.44
max	16.14	20.88

4.2 Boxplot



4.3 Normality Test

```
Global Temperature:
W_stat: 0.9990
P-value: 0.7460
Normal data

Sea Surface Temperature:
W_stat: 0.9985
P-value: 0.4022
Normal data
```

4.4 F-Test

```
f_stat = 0.1635
p_value = 0.0000
equal variances
```

4.5 2-Sample t-Test

```
t_stat = -43.2321
p_value = 0.0000


Global average temperature
is significantly different
from sea surface temperature.
```

4.6 Power of the Test

```
Cohen's d: 0.8529

Power of the test:
100.00%
```

5. Conclusion

A two-sample t-test was conducted to evaluate whether global average temperature differs significantly from sea surface temperature. Based on the computed t-statistic and corresponding p-value from the analysis, the results indicate whether the null hypothesis should be rejected at the 0.05 significance level. If the p-value is less than 0.05, the null hypothesis is rejected, suggesting that there is a statistically significant difference between global average temperature and sea surface temperature.